

Layers

Read the following sentence.

The horse raced past the barn fell.

FIRST READ

It doesn't make sense. Did someone forget a word?

READ IT AGAIN

Wait, did a barn fall? Did the horse race past the barn afterward?

READ IT A THIRD TIME

Got it. A horse ran past a barn. And after running past the barn, the horse fell.

Each pass, your mind updates the meaning until it clicks.

AI DOES THE SAME THING

It reads your message over and over, establishing the meaning a little more with each pass. And because language is full of nuance, it takes dozens of passes to pin the meaning down.

Each pass is called a **layer**, and at every layer AI runs the two moves you just met: **attention** (which words matter) and **transformation** (update the meaning).

And AI does it with math. At each layer, it adjusts the tokens' numbers, each pass moving them a little closer to what the tokens mean.

THE MECHANICS

You saw this sentence in the Transformer lesson. Here's how the numbers adjust for one word: **IT**. Notice the last box is blank. That box gets filled in the next lesson.

Thecat sat on the mat during the May rainstorm because it was

tired

AMBIGUOUS "IT"

On its own, **IT** is just a pronoun. Its vector could mean almost anything, and certainly not **CAT**.

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THROUGH THE LAYERS

Each layer reads the surrounding words and shifts the numbers for **IT**. Layer by layer, the model works out what **IT** refers to.

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What does this look like inside the model? Like this.

What happens inside each layer

Each layer refines every token's vector, then hands it to the next layer.

LAYER 1

STARTING VECTOR

[0.42, -1.15, 0.33, 2.08, -0.73, ...]

ATTENTION

Which words matter?

TRANSFORMATION

Update the meaning

RICHER VECTOR OUT

[0.51, -0.87, 0.21, 1.91, -0.26, ...]

LAYER 2

FROM LAYER 1

[0.51, -0.87, 0.21, 1.91, -0.26, ...]

ATTENTION

Which words matter?

TRANSFORMATION

Update the meaning

RICHER VECTOR OUT

[0.27, -1.21, 0.67, 2.24, 0.14, ...]

LAYER 3

FROM LAYER 2

[0.27, -1.21, 0.67, 2.24, 0.14, ...]

ATTENTION

Which words matter?

TRANSFORMATION

Update the meaning

RICHER VECTOR OUT

[0.31, -0.92, 0.48, 1.74, -0.31, ...]

DOZENS MORE LAYERS

FROM LAYER 3

[0.31, -0.92, 0.48, 1.74, -0.31, ...]

ATTENTION

Which words matter?

TRANSFORMATION

Update the meaning

RICHER VECTOR OUT

[0.19, -1.12, 0.72, 2.13, -0.03, ...]

Same two moves at every layer. Each pass, the numbers move closer to what the token means.

WHY ARE THERE DOZENS OF LAYERS?

Because some meaning is many steps deep.

A FEW PASSES

Simple meaning resolves early. The plain sense of a sentence is settled in a handful of passes.

DOZENS OF PASSES

Catching sarcasm, following a twist in a story, or reasoning through a complicated problem takes many more.

WHY NOT HUNDREDS?

Past a point, extra depth stops helping. It just makes the model more expensive to run.

A few layers reach only shallow meaning. Stacking dozens leaves room for the deep kind.

NEURAL NETWORKS

The whole stack of layers is called a **neural network**. The design is loosely borrowed from biology, simple units passing signals forward the way neurons do. But that's where the resemblance ends. It isn't a brain and it isn't thinking: it's the same arithmetic, repeated billions of times, fast.

Meaning builds up, layer by layer.

The blank box gets filled in the next lesson.